

Afzal Hussain

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EDUCATION

Islamic University of Technology 2018 B.S., Mechanical Engineering; CGPA: **3.70**/4.00
IUT Gazipur, Bangladesh

RESEARCH EXPERIENCE

My research interests are scientific machine learning, computational mechanics, machine learning-accelerated CFD, engineering design optimization, simulation-driven design exploration, and AI-assisted engineering decision-making.

Self-Directed Research

Computational Fluid Dynamics

Image-Based CFD Using Deep Learning

Built a deep-learning surrogate for transient compressible CFD using approximately 1,500 OpenFOAM simulations of Mach-3 flow over parameterized forward-facing-step geometries. Developed the simulation, preprocessing, training, and visualization pipeline using a ConvLSTM-ResNet architecture to predict velocity, pressure, and temperature fields from prior flow states.

<https://github.com/afzalhussain23/>

[Image-Based-CFD-Using-Deep-Learning](#)

Islamic University of Technology

Gazipur, Bangladesh

Undergraduate Thesis

Advisor: Prof. Mohammad Ahsan Habib

Developed a hybrid Whale Optimization Algorithm for multi-objective engineering optimization in stainless-steel turning.

The work was peer-reviewed and published in the *Journal of Manufacturing and Materials Processing*.

PUBLICATIONS

- Multi-Objective Optimization of Turning Operation of Stainless Steel Using a Hybrid Whale Optimization Algorithm. Mahamudul Hasan Tanvir, **Afzal Hussain**, M. M. Towfiqur Rahman, Sakib Ishraq, Khandoker Zishan, SK Tashowar Tanzim Rahul, and Mohammad Ahsan Habib. *Journal of Manufacturing and Materials Processing*, 4(3), 64, 2020. <https://doi.org/10.3390/jmmp4030064>
- Effectiveness of Hard Clustering Algorithms for Securing Cyber Space. Sakib Mahtab Khandaker, **Afzal Hussain**, and Mohiuddin Ahmed. In *Smart Grid and Internet of Things*, Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering, pp. 113–120. Springer International Publishing, 2019. https://doi.org/10.1007/978-3-030-05928-6_11

INDUSTRY EXPERIENCE

Pathao (Dhaka, Bangladesh)	03/2023 – 11/2025	Senior Software Engineer Led modernization of large-scale distributed systems serving over 10 million users, with emphasis on scalable computation, reliable workflows, and data-intensive software. Built durable asynchronous systems for failure-prone processes and explored LLM-assisted engineering workflows; this strengthened my ability to develop robust computational tools for simulation, optimization, and AI-assisted engineering design.
Hogarth (Dhaka, Bangladesh)	11/2021 – 02/2023	Software Engineer Developed enterprise software systems for global clients across multiple deployment environments, gaining experience in modular design, testing, debugging, and collaborative development. Led investigations of mission-critical production issues using structured root-cause analysis and careful validation.
Enosis Solutions (Dhaka, Bangladesh)	12/2018 – 10/2021	Software Engineer Built database-backed modules for point-of-sale, order fulfillment, inventory synchronization, reporting, and real-time communication systems. Developed a foundation in data modeling, computational workflow design, and reliable end-to-end software implementation.

AWARDS AND HONORS

- Ranked **18th** nationally in the Bangladesh Olympiad in Informatics, 2012 and 2014.
- Ranked **2nd** in IUT intra-university competitive programming team selection; **Best Freshman Team** performance from IUT at ACM-ICPC Dhaka Regional, 2015.
- **National finalist**, Bangladesh Physics Olympiad and Bangladesh Astronomy Olympiad, 2013.

LEADERSHIP AND SERVICE

- Coached university students in algorithms and data structures, preparing them for competitive programming contests.
- Executive Member, IUT Computer Society; helped organize technical activities and student programming initiatives.
- Provided technical leadership for software engineers through code reviews, onboarding, architecture discussions, and distributed-systems design reviews.

TECHNICAL SKILLS

- Programming and scripting: Python, C/C++, MATLAB, SQL, Bash/Linux.
- Machine learning and scientific computing: PyTorch, TensorFlow, Keras, NumPy, SciPy, pandas, scikit-learn, Matplotlib, Jupyter Notebook.
- Language models: Transformer architectures, attention mechanisms, tokenization, language-model implementation and evaluation, LLM-assisted development.
- Engineering simulation and design: OpenFOAM, computational fluid dynamics (CFD), computer-aided design (CAD), engineering optimization.
- Software systems: distributed systems, event-driven architecture, observability, production reliability, reproducible data pipelines, database-backed systems.
- Infrastructure and data: Docker, Kubernetes, Amazon Web Services (AWS), PostgreSQL, Git.

SELECTED COURSEWORK

- Language Modeling from Scratch (CS336), Stanford University, in progress.
- Deep Learning Specialization, Coursera.
- Machine Learning, Coursera.
- A Hands-on Introduction to Engineering Simulations, edX.
- Using Python for Research, edX.